

GFO-Light: A Simplified Top-Level Ontology — Introduction and Biomedical Case Studies

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Abstract. We present a novel lightweight top-level ontology (TLO) ‘GFO-light’, which has been designed for ease of use by non-ontologists. While most TLOs come with rich, complex axiomatizations and terminologies that include a lot of philosophical jargon, lightweight TLOs aim to strike a balance between expressivity and accessibility by domain experts without excessive expertise in formal ontology. GFO-light is derived from the General Formal Ontology (GFO), which has unique features, such as the integration of objects and processes, and coincidence in continuous spacetime. The ontological and representational choices guiding the construction of GFO-light were driven by different real-world use cases and by the design choice to accommodate as many parts of the original GFO ontology as possible, as well as common modeling patterns. The novel GFO-light TLO attempts to strike a balance between ease of use, expressiveness, and inclusion of widely used features lacking from other lightweight TLOs.

Keywords. Classification, Terminology, Semantics, Interoperability

1. Introduction

Biomedical domain ontologies, such as the Gene Ontology, the Foundational Model of Anatomy, and the Disease Ontology have become essential tools in biomedical informatics, providing structured representations of knowledge that facilitate tasks such as data standardization and integration, automated reasoning, and knowledge discovery. However, domain ontologies often exhibit inconsistencies in terminology, structure, and conceptualization, making it challenging to integrate information across multiple domains. These challenges can be addressed by means of top-level ontologies (TLOs), which provide a high-level, domain-independent conceptualization of fundamental categories (e.g., objects, processes, and roles) that serves as a unifying foundation for structuring and interrelating domain ontologies, ensuring cross-domain logical

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consistency and semantic interoperability, and thereby facilitating meaningful data exchange and integration in complex biomedical systems. Despite their benefits, TLOs are not widely adopted in domain ontology development, likely due to high adoption costs, strong ontological commitments, and the need for rigorous training. Additionally, OWL has become the de facto standard representation formalism for domain ontologies, whereas most TLOs use more expressive systems like first- or higher-order logic, due to their high level of formalization. To improve usability, some TLO creators have released lightweight versions that reduce ontological commitments and formal complexity, making them easier to handle and compatible with common ontology languages like OWL. Two prominent TLOs that have a 'light' counterpart are, at the time of writing, DOLCE with DOLCE-lite and DOLCE Ultralite [1], and UFO with its 'gentle' gUFO [2].

As core contributions of our work, we have derived the novel lightweight TLO 'GFO-light' in the context of the General Formal Ontology and its applications, aiming at the right balance between ease of use, degree of formalization and sufficient conceptual expressiveness for common cases of building biomedical domain ontologies and applications around them. Its development is intertwined with real-world projects, which guided its advancement and first evaluation.

2. Methods

Since its inception, the General Formal Ontology (GFO) has been applied in various industrial use cases and research projects, including 1) a core ontology for biology, 2) ontology-based search, 3) risk assessment and avoidance of adverse events, 4) computer-aided navigation in minimally invasive surgery, 5) metadata repositories for clinical and epidemiological research, 6) a core ontology for anthropology, 7) data discovery and insights generation in the context of the SMITH project, 8) function modeling, 9) cell tracking and time-lapse microscopy, and 10) process modeling [3]. These applications not only validated and shaped GFO's evolution but also highlighted distinct conceptual areas of its theory. These include, among others (with numbers of the corresponding use cases): categories of different orders and individuals (1, 5, 7); symbolic structures and denoted concepts (2); processes and situations (3, 4, 10); objects, their parts and aggregates (4, 6, 7, 9); attributives and functions (3, 5, 6, 7, 8); spatial and temporal entities (3, 6, 9). The observation that different applications require distinct subsets of GFO has driven its modularization. Accordingly, current development focuses on a modular structure, which will be implemented in the upcoming GFO version 2.0 [4].

Two further insights emerged: 1) Despite relying on different subsets of GFO, all applications share a common conceptual core forming the basis for a core ontology (GFO-core). 2) However, some use cases require a broader (cross-module) view but, in turn, less ontological depth (GFO-light). By analyzing the various GFO use cases, we identified a set of common entities. Additionally, we examined common modeling patterns, especially for temporal relations, and adapted ontological terminology to better align with domain experts' practices. For example, while GFO and other TLOs provide a formal treatment of time, they lack intuitive relations between temporal entities, such as 'before' or 'after', requiring domain experts to define them on top of the TLO's time theory following the usual design patterns. To alleviate users from this recurring task, GFO-light includes ready-to use time relations. Modeling patterns were adopted based on the existence of corresponding standards. Next, the design principles common to existing lightweight TLOs were applied. The aim was to reduce complexity and

ontological commitment to such a degree that it can be entirely represented in OWL while maintaining the possible maximum of expressiveness and access to unique features of GFO, such as Brentano time. Finally, we validated GFO-light based on the following three biomedical use cases and compared it with other lightweight TLOs. These use cases are: 1. *the Anthropological Notation Ontology (ANNO)* [5], which allows for the systematic description and classification of recovered bone finds in anthropology, 2. *the Risk Identification Ontology (RIO)* [6], which provides a framework for risk specification and reasoning, and 3. *the Core Ontology of Phenotyping (COP)* [7], which defines core entities that are relevant for modeling phenotypic knowledge and developing phenotyping software.

3. Results

The result of the approaches described in Section 2 is GFO-light, a simplified version of GFO, which contains the following core categories (see Figure 1 and the online documentation²): **Continuants** are concrete individuals that persist through time and are wholly present at every point in time of their existence. Social or material (three-dimensional) objects, their groups/aggregates, parts and boundaries (two-, one- and zero-dimensional entities, i.e., surfaces, lines and points) are subsumed under the category Continuant. **Attributives** are concrete individuals that depend on other individuals by some kind of dependency relation. Qualities (traits, characteristics, etc.) of individuals, relations (relators) between them and roles that individuals play in different contexts are subsumed under the category Attributive. **Processes** are concrete individuals that happen in time and have a temporal extension (time interval). Processes can have parts, which are situations, attributives or further processes. **Situations** are combinations of contextually related attributives. A situation can exist at a certain point or period in time and can be part of a process or another situation. GFO-light distinguishes between two time categories, **Chronoid** (time interval) and **Time Boundary** (point in time), which are used, among other things, to define the time at which concrete individuals exist.

GFO-light is fully sufficient for modelling the use cases introduced in Section 2. In the ANNO use case, anthropological entities are represented as subclasses of Continuant. Their compositions (partOf), aggregations (memberOf) and spatial relations to each other are modelled using Relator with target and reference object roles. Phenotypes and other properties are modelled as subclasses of Attributive. In comparison to full GFO, GFO-light adds a distinction between Object, ObjectAggregate, and ObjectPart, enabling a simplified and clear modeling of composition and aggregation. It also allows the roles of a relator to be modelled as OWL object properties instead of defining them as instances of the class Role.

The RIO use case brought about a new interpretation of the notion of situation: In the original RIO, the properties (including risk properties) of the entities involved in treatment situations (patients, doctors, instruments, etc.) were artificially considered as properties of the situations themselves (gfo:has_property). GFO-light (and thus the new RIO) offers a more natural view of a situation as a combination of properties (Attributive), i.e., a situation consists of (hasSituationPart) attributives. An example of a risk situation related to cochlear implantation (CI) would be a situation consisting of the following attributives: patient age = 2 months, meningitis vaccination = false, and

² <https://github.com/On-to-Med/gfo-light/blob/main/docs/README.md>

antibiotic prevention = false. Another simplification is that, in GFO-light, points in time of a process can be assigned directly to the process itself (temporalPartOf), without a detour via its temporal extension (Chronoid). In addition, some GFO-light object properties (e.g., processPartOf, situationPartOf, and leadsTo/resultsFrom) could be reused, meaning it was no longer necessary to define similar properties in RIO.

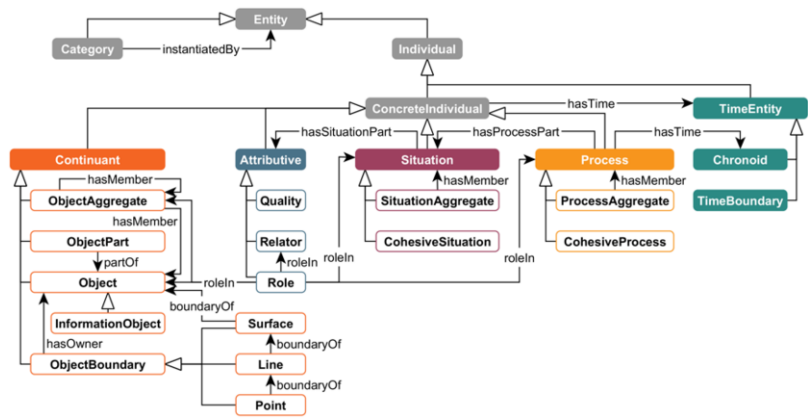


Figure 1. The conceptual structure of GFO-light.

COP distinguishes between phenotypes, i.e., individual traits or characteristics (Attributive) of an organism (Object) and phenotype classes (Category), which are instantiated by phenotypes. Examples of phenotype classes are (abstract) categories such as ‘weight’, ‘overweight’, or ‘body mass index (BMI)’, which are instantiated by the weight, overweight, and BMI properties (attributives) of specific individuals. Furthermore, GFO-light allows the definition of information objects that specify or describe other entities. Therefore, Algorithm, Model, and Software are subcategories of InformationObject in COP. The phenotyping is considered as a process (Process) of identifying individuals (Object) with certain phenotypes. A run of a phenotyping software, which implements a phenotyping algorithm, realizes the phenotyping.

4. Discussion

As pointed out in Section 3, GFO-light is the only lightweight TLO incorporating common modeling patterns for temporal relations. Therefore, the representation of a temporal order between treatment phases, as well as the restriction that all phases of the treatment must be temporally included in the overarching treatment process, is only possible in GFO-light. DOLCE Ultralight permits the representation of a temporal order of situations via the introduction of so-called transition events. The use of transition events, however, implies not only a temporal but also a causal relation between the connected processes, which is not always desired. Moreover, in many use cases (as, for example, in the RIO use case), only the sequence of processes itself is of interest. COP distinguishes between Phenotype Class as a category and Phenotype as its instances. This distinction is only possible in GFO-light and gUFO (gUFO’s equivalent of Category being ‘Type’), while DOLCE Ultralite does not include categories. While GFO-light and DOLCE Ultralite include information objects, it is unclear how to represent this type of immaterial object in gUFO, which does not make an explicit distinction between

symbolic and other kinds of objects. Balancing expressivity with lightweight design, GFO-light omits certain entities, which might require alternative top-level ontologies for complex use cases. One prominent example is the absence of presentials in GFO-light, which are snapshots of continuants and the basis of GFO-full's integration principle of objects and processes [8]. A scenario where GFO-light's expressiveness is limited by this omission is the direct modeling of timeline-based evolutionary processes, such as cell tracking in time-lapse microscopy. However, time snapshots can be represented as situations without temporal extension (GFO-light allows situations to exist over time intervals and at singular points in time). Future work will consist in the evaluation of GFO-light's applicability to a broader set of usage scenarios and modeling problems. In upcoming versions of GFO-light, we will focus on the exploitation of more of the semantic features of OWL, especially characteristics of relations (such as transitivity, symmetry, and reflexivity).

5. Conclusion

We have introduced the novel lightweight top-level ontology GFO-light. It was created in the context of the broader development process of the General Formal Ontology (GFO), which meanwhile aims at a modular framework called GFO 2.0 [4]. Several scenarios have been presented that motivated the development and demonstrated the applicability of GFO-light. In a comparative analysis including state of the art lightweight TLOs, the advantages and trade-offs of GFO-light were discussed.

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